

Using Earables Platforms to Study Verbal Communication

Introducing earables to psycholinguistic research

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ABSTRACT

Earables provide a new opportunity to study conversation in the wild. They uniquely allow (i) accurate head motion tracking recorded synchronously with the speech signal and (ii) multiple people to simultaneously receive and stream conversational speech that is unconstrained by body movement. Here, our general aim is to introduce the use of earables for conducting psycholinguistic studies requiring audio and movement data jointly collected during verbal interaction in a natural setting. Specifically, we propose using earables platforms to address the relationship between head movement, speech and meaning transmission from single and multiple-person perspectives.

CCS CONCEPTS

• **Applied computing** → Law, social and behavioral sciences; Psychology; • **General and reference** → Document types; General conference proceedings.

KEYWORDS

Earables, head motion, conversation, linguistics, interpersonal, coordination, turn-taking, speech, verbal communication

ACM Reference Format:

Alejandro Pérez and Matthew H. Davis. 2022. Using Earables Platforms to Study Verbal Communication: Introducing earables to psycholinguistic research. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3544793.3563414>

1 PROCEDURE

We will collect accelerations, rotational velocities and audio signals with the eSense platform [1, 2] to measure head movements and turn-taking length during structured conversations. Data will be recorded while two participants play the board game “Taboo”. This way, we can obtain concurrent linguistic measurements, along with a quantitative indicator of the success of an interaction, like the number of words correctly guessed, the number of errors and the number of conversational turns or the amount of time required

to generate these guesses. Moreover, the task allows a similarly structured conversation across multiple participant dyads and the ludic aspect of the activity act as a prompt to facilitate the interaction. Potentially, we can include several groups, each of them using a different language. Lab Streaming Layer (LSL) software will synchronise the recordings across multiple participants. The relationship between signals will be measured (in bites) by computing the mutual information (MI) between time series (e.g., Gaussian Copula MI). Speech signals will be automatically transcribed to text (e.g., Google Cloud Speech-to-Text API), including speaker diarisation and words’ time stamps to measure the turn-taking length. The entropy/predictability for each word and speaker intervention will be obtained by applying natural language processing (NLP) techniques (e.g., GPT-3 model).

1.1 Studying the relationship between head movements and speech

Head movements and verbal communication are interwoven; an inherent kinematic component is associated with speech production. Furthermore, by moving the head, we can transmit information (e.g., nods of agreement for approval). Tongue and mandibular movements provide visual cues that facilitate comprehension and interlocutor engagement [3]. These movements strongly track with the speech envelope. In turn, speech envelope information is essential for language comprehension and is closely tracked by listeners’ brain responses to achieve speech intelligibility [4].

Earables offer considerable potential for quantitatively exploring head movement’s communicative and linguistic function [5]. Here we will measure the mutual relationships between speech envelope and head motion. In recent work testing the motion tracking in the eSense platform, Ferlini et al. [6] describe that “speaking generates unwanted vibrations and micro-movements that are captured by the sensors”. We hypothesise that these micro-movements captured by earables are associated with the speech envelope. Thus, speakers’ (and listeners’) head motion tracking of speech envelope will be measured. To test for a significance requirement of 95%, 1000 surrogates will be created for each participant. These surrogates will be obtained by perturbing the speech envelopes while preserving autocorrelation. The perturbed speech envelopes will be paired with intact head motion data, and the GCMI calculated. Then, we will compare if the group mean GCMI value is larger than the maximum null mean GCMI. In summary, we will identify whether the speech envelope provides information regarding head motion. If it is true that head micro-movements detected by the earables are correlated to the speech envelope, they could be removed from the motion signal and used as a backchannel to increase speech engagement in the listeners of the produced speech.

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UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom
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ACM ISBN 978-1-4503-9423-9/22/09.
<https://doi.org/10.1145/3544793.3563414>

1.2 Studying interpersonal coordination during a conversation

Interpersonal non-verbal synchronization and smooth conversational turn-taking (i.e., with short gaps) have both been associated with successful face-to-face verbal interactions [7]. We propose to explore this association further using earables. Specifically, we will test whether head movement signals from two or more interlocutors can be used to measure the moment-by-moment degree of interpersonal synchronization. Moreover, we will test whether interpersonal synchronization correlates with the magnitude of the turn-taking lag. Finally, we will use these measurements to predict the “successfulness” of verbal exchange.

To test the validity of using earables motion signals to capture interpersonal synchronization, we will compare the two-person head motion mutual dependence against surrogate data created by re-pairing members of different, non-interacting dyads. Using the same measure, we will also evaluate the performance of the automatic classification of pairs of participants engaged in an interactive conversation as distinct from the randomly paired participants. Then, we will correlate these common head movement patterns with the gap length preceding the switch between listening and speaking. Finally, measures of the two heads’ movement relationship and the length between conversational turns will be used to predict measures quantifying the efficacy of the verbal information exchange during the conversation. In summary, we will quantify the relationship between interpersonal coordination and the success of meaning transmission between conversational partners.

2 APPLICATIONS AND SIGNIFICANCE

Conversational turn-taking is a critical type of coordination necessary for conversation. At any moment, there is a speaker and a listener, whose roles alternate to produce the two-way meaning transmission that characterizes a dialogue. We sense when to jump into the conversation, aided by grammatical cues, intonation and the ‘backchannel’ responses provided by head movements. For example, head movements can emphasize chunks of speech that are supposed to be uninterrupted [8]. However, there are situations where interlocutors have limited or no concurrent visual information. This can be the case with online interactions like avatar chats, in-game dialogues and telephone conversations. Through earables, we can synthesize spontaneous head movements to be delivered as a surrogate backchannel signal (e.g., vibrations) to help coordinate interlocutors’ timing, supporting better conversational outcomes.

From head movement signals, we can determine which interpersonal interactions are most important for a given individual. This information can reveal which signals to prioritize in a multi-person situation to guide a directional hearing aid. Although there are other vision-based methods that are more reliable for head motion monitoring, earables are discrete and socially accepted, and allow real-life conversational recordings minimally supervised. We might also devise systems for providing feedback or training to individuals that struggle in everyday social situations, for example, by detecting poor timing in their conversational interventions or head movements that are at odds with prevailing cultural norms. Such systems might, in time, prove valuable for facilitating communication and enhancing speech comprehension.

We anticipate that measurements obtained from earables can reveal important cues to the status and quality of ongoing vocal communication. By combining interpersonal synchronization, turn-taking and speech predictability structure from NLP models, we will be able to tag and online monitor the nature of the interaction (e.g., socializing vs arguing) and the degree of bonding (empathy) between interlocutors.

In a nutshell, earables platforms enable the low-cost, unobtrusive monitoring of multiple people during prolonged naturalistic interaction. Altogether, allow researchers to conduct real-life experiments on interpersonal verbal communication. Here we emphasize that earables facilitate investigating how participants coordinate during verbal interaction.

ACKNOWLEDGMENTS

AP received funding from the European Union’s Horizon 2020 Research and Innovation Programme under the Marie Skłodowska-Curie Individual Fellowship, grant agreement No 840885. MHD received intramural funding from the Medical Research Council (MC_UU_0005/5).

REFERENCES

- [1] Kawsar F, Min C, Mathur A and Montanari A, 2018. Earables for Personal-scale Behaviour Analytics. *IEEE Pervasive Computing*, Volume: 17, Issue: 3.
- [2] Min C, Mathur A, and Kawsar F, 2018. Exploring Audio and Kinetic Sensing on Earable Devices. In *WearSys 2018, The 16th ACM Conference on Mobile Systems, Applications, and Services (MobiSys 2018)*, Munich.
- [3] Medina S, Tome D, Stoll C, Tiede M, Munhall K, Hauptmann AG, Matthews I, 2022. Speech Driven Tongue Animation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 20406-20416.
- [4] Peelle JE, Gross J, Davis MH, 2013. Phase-locked responses to speech in human auditory cortex are enhanced during comprehension. *Cereb. Cortex*, 23(6), 1378-87.
- [5] McClave EZ, 2000. Linguistic functions of head movements in the context of speech. *J. Pragmat.* 32, 855–878.
- [6] Ferlini A, Montanari A, Mascolo C, Harle R, 2019. Head Motion Tracking Through in-Ear Wearables. In *Proceedings of the 1st International Workshop on Earable Computing (EarComp 2019)*, pp. 8-13.
- [7] Lahnakoski JM, Forbes PAG, McCall C, Schilbach L, 2020. Unobtrusive tracking of interpersonal orienting and distance predicts the subjective quality of social interactions. *R. Soc. Open Sci.* 7: 191815.
- [8] Shockley K, Santana M-V, Fowler CA, 2003. Mutual interpersonal postural constraints are involved in cooperative conversation. *J. Exp. Psychol. Hum. Percept. Perform.* 29(2), 326–332.